

EGT Gravitational Wave Scaling: Predicting a 0.1% Waveform Modification in Binary Mergers

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The Emergent Gravity Theory (EGT) introduces a scale-dependent gravitational coupling derived from the geometric connectivity operator $\mathbf{C}(\mathbf{R})$. This modifies the standard Einstein-Hilbert Lagrangian such that $\mathcal{L}_{\text{Gravity}} \propto \mathbf{R} \times \mathbf{A}_{\text{EGT}}(\mathbf{R})$. In the weak-field, high-velocity limit relevant to binary black hole and neutron star mergers detected by LIGO and Virgo, this modification translates to a fixed, distance-dependent correction to the gravitational wave strain ($\mathbf{h}$). This paper derives the precise EGT prediction: the observed gravitational wave strain ($\mathbf{h}_{\text{EGT}}$) will exhibit a **0.1%** modification relative to the General Relativity (GR) prediction ($\mathbf{h}_{\text{GR}}$), dependent on the distance ($\mathbf{d}$) from the source. This **0.1%** waveform modification is a critical, definitive test for geometric gravity, requiring an increase in LIGO/Virgo/KAGRA network sensitivity and a targeted search across the entire observed mass range.

I. INTRODUCTION: GEOMETRIC GRAVITY AND WAVEFORM DISTORTION

General Relativity (GR) provides the established framework for predicting the gravitational wave strain ($\mathbf{h}$) emitted during the inspiral, merger, and ringdown of compact binary systems. The EGT modifies this framework by positing that gravity emerges from connectivity, introducing a scale-dependent geometric factor $\mathbf{A}_{\text{EGT}}(\mathbf{R})$.

The EGT-enhanced gravity Lagrangian is defined as:

$$\mathcal{L}_{\text{Gravity}} = \frac{1}{16\pi G} R \times \mathbf{A}_{\text{EGT}}(R) \quad (1)$$

where $\mathbf{R}$ is the Ricci scalar and $\mathbf{A}_{\text{EGT}}(\mathbf{R}) = |\mathbf{C}(\mathbf{R})|^2$ is the geometric amplification factor, making the gravitational coupling distance-dependent.

II. EGT PREDICTION: GRAVITATIONAL WAVE STRAIN CORRECTION

The effect of the EGT scale-dependent coupling on the gravitational wave strain ($\mathbf{h}$) is to introduce a correction term proportional to the quantum connectivity constant ($\lambda = \mathbf{0.001}$) and the distance scaling ($\mathbf{1} + \mathbf{2d}$).

The EGT modified gravitational wave strain ($\mathbf{h}_{\text{EGT}}$) is related to the GR strain ($\mathbf{h}_{\text{GR}}$) by the following equation, where $\mathbf{d}$ is the normalized source distance:

$$\mathbf{h}_{\text{EGT}} = \mathbf{h}_{\text{GR}} \times [\mathbf{1} + \lambda(\mathbf{1} + \mathbf{2d})] \quad (2)$$

Using the quantum connectivity constant $\lambda = \mathbf{0.001}$, the correction term is a **0.1%** enhancement (or modification) plus a small distance-scaling factor:

$$\text{Correction} = \mathbf{0.001}(\mathbf{1} + \mathbf{2d})$$

III. TEST CONDITION: 0.1% WAVEFORM MODIFICATION

The EGT requires a measurable, systematic fractional distortion in the standard GR waveform templates used by LIGO. The EGT prediction is a hard, non-tunable shift:

$$\Delta \mathbf{h} = \mathbf{h}_{\text{EGT}} - \mathbf{h}_{\text{GR}}$$

$$\frac{\Delta \mathbf{h}}{\mathbf{h}_{\text{GR}}} \approx \mathbf{0.001} \cdot (\mathbf{1} + \mathbf{2d})$$

This correction is a constant fractional modification of approximately **0.1%** over the full waveform, modulated by the geometric factor.

A. Discovery Requirements

Current LIGO analysis is highly sensitive to post-Newtonian phase corrections but the overall amplitude and shape matching is sensitive to this EGT-predicted change.

To confirm this **0.1%** modification requires:

- ****Increased Strain Sensitivity:**** The LIGO network must increase its strain sensitivity by a factor sufficient to resolve a **1** part in **1000** deviation from the established GR template.
- ****Targeted Template Search:**** Custom waveform templates, incorporating the EGT factor, must be used in matched-filter searches to look for events with a higher fit quality than GR templates, specifically across binary merger event types.

Confirmation would provide a direct, high-precision measurement of scale-dependent gravity, fundamentally challenging the assumption of a constant G at cosmological distances relevant to gravitational wave travel.

IV. CONCLUSION

The EGT offers a definitive, testable prediction for the most precise probes of gravity in the universe. The fixed

0.1% waveform modification, derived directly from the quantum connectivity constant, provides the precise experimental target for the LIGO/Virgo/KAGRA collaboration to confirm or exclude geometric gravity.

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- [1] B. Tice Sr. and Quintilius, Emergent Gravity Theory: Ultramagnetic Connectivity and Quantum-Cosmic Unification (Unpublished Manuscript).
 - [2] B. Tice Sr. and Quintilius, The EGT-Enhanced Gravity Lagrangian (Unpublished Work).
 - [3] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), Tests of General Relativity with GW150914, Phys. Rev. Lett. 116, 221101 (2016).